# AN EXACT EDGE-COUNTING IDENTITY FOR THE GALLAI-EDMONDS PARAMETERS OF A GRAPH

SAHAR DISKIN, ANNA GEISLER

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*Allowed standard prerequisites (editorial).*

- The Gallai-Edmonds decomposition (D, A, C) of a graph, its parameters r and h, and the components D_i of G[D]. Set up in the author's subsection on the Edmonds-Gallai decomposition (source lines 1054-1098), where the parameters r and h are defined. Fixes every symbol that appears in the identity, in particular the sets A, C, D, their sizes a, r, and the quantity h.

**Lemma 0.1.**

$$2h = dr + 2e_G(A) + e_G(A \cup D, C) + \sum_{i:|D_i|\geq 3} \big(e_G(D_i, D_i^c) - d\big). \tag{1}$$

*Proof.* Degree summation in $D$ gives

$$d|D| = 2\sum_i e_G(D_i) + 2\sum_{i<j} e_G(D_i, D_j) + e_G(D, A) + e_G(D, C).$$

Rearranging for $h = \sum_{i<j} e_G(D_i, D_j) + e_G(D, C)$, we obtain

$$2h = d|D| - 2\sum_i e_G(D_i) - e_G(D, A) + e_G(D, C). \tag{2}$$

Similarly, degree summation in $A$ gives

$$d|A| = 2e_G(A) + e_G(A, D) + e_G(A, C). \tag{3}$$

Subtracting the LHS and adding the RHS of (3) to (2), we obtain

$$2h = d(|D| - |A|) - 2\sum_i e_G(D_i) + 2e_G(A) + e_G(A, C) + e_G(D, C).$$

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Now

$$|D| - |A| = r + \sum_{|D_i|\geq 3} (|D_i| - 1), \qquad 2e_G(D_i) = d|D_i| - e_G(D_i, D_i^c),$$

and since each $|D_i|$ is odd both sums over $i$ may be combined, proving the claim. □